

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. (ME) Examination

December 2012

ME 713.01 Advanced Materials

Date: 18.12.2012, Tuesday Time: 10:00 a.m. To 01:00 p.m. Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Classify characterization techniques of nanostructured materials. Discuss any three applications of thin films in detail. [07]

- Q - 2 (a)** Explain stages of thin film fabrication, its scale up and manufacturability in detail [07]
(b) What are the differences between evaporation and sputtering? Explain sputtering process in detail. [07]

OR

- Q - 2 (a)** Write short note on Chemical Vapor Deposition (CVD) process. [07]
(b) Explain various growth processes of nanostructured thin films in detail. [07]

- Q - 3** Attempt any two [14]
(a) Write short note on dislocation models for films strength.
(b) Write short note on hardness and strength of multilayer films.
(c) Explain in detail thermal stress in homogeneous, bilayer and three-layer structures.

SECTION – II

Q - 4 Discuss in detail applications of advanced nanostructured materials in areas such as coating, transparent conductive thin films, solar cells, cutting tools & flat panel displays. [07]

- Q - 5 (a)** Explain reflection, absorption, transmission, color, opacity and translucency in detail. [07]
(b) Explain substrate deflection and x-ray diffraction techniques for measuring internal stress in films. [07]

OR

- Q - 5 (a)** Explain applications of optical phenomena such as luminescence, photoconductivity, and lasers in detail. [07]
(b) Explain intrinsic stress in thin films and dependence of stress on sputtering variables. [07]

- Q - 6** Attempt any two [14]
(a) Explain contact angle determination by sessile drop.
(b) Explain water drop contact angle measurements on substrates.
(c) Write short note on frontier areas for applications and selection criteria of nanostructures.
